

Dynamical-Spacetime Discrete-Ordinates Radiation in AthenaK

Circumbinary radiation implementation draft

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Abstract

We describe a standalone discrete-ordinates radiation module for AthenaK that can evolve radiation on time-dependent ADM backgrounds. The existing radiation solver is tied to stationary Cartesian Kerr-Schild metrics: it stores a Killing-energy weighted intensity, uses metric-dependent angular drift coefficients that can be precomputed, and is therefore efficient for isolated stationary black holes. These assumptions are not appropriate for binary black hole backgrounds or for metric data supplied by a dynamical spacetime solver. The new `dyn_radiation` module keeps an exact compatibility mode for the legacy stationary solver, but adds an ADM formulation in which the evolved quantity is the Eulerian densitized intensity, the spatial transport speed is $\alpha s^i - \beta^i$, angular geodesic advection is computed from the ADM Hamiltonian, and the geometric energy source is written in Valencia form using α , β^i , γ_{ij} , and K_{ij} . This avoids assuming a time-independent metric and does not require explicit gauge-field time derivatives beyond the ADM metric-evolution relation for $\partial_t \gamma_{ij}$. In the present implementation the radiation field is metric-passive: it is transported on ADM/Z4c geometry, but its stress-energy is not included in the Z4c metric right-hand side. We also add a diagnostic null-geodesic particle pusher for beam-edge tracers on ADM backgrounds. Flat-space beam tests reproduce the legacy solver to roundoff, radiation-hydrodynamic linear-wave tests reproduce the legacy error norms in stationary Kerr-Schild mode, and a superposed boosted-Kerr-Schild binary test exercises the ADM beam source, ADM matter coupling, and null-particle tracers.

Keywords: numerical relativity; radiation hydrodynamics; general relativistic radiation transport; binary black holes; AthenaK

1 Introduction

Radiation transport in relativistic accretion flows is usually implemented on a fixed analytic background. For a single Kerr black hole this is a good approximation in many applications: the metric is stationary, the coordinate tetrad can be generated once, and the angular advection terms that bend rays in momentum space can be precomputed. Circumbinary and dynamical-spacetime applications remove these simplifications. The lapse, shift, spatial metric, and extrinsic curvature may change in time; the background may be supplied by a numerical relativity evolution; and the same radiation module should be usable with superposed black-hole data, Z4c data, or any other ADM source.

The goal of the implementation described here is narrower than a full production radiation scheme. We add a separate `dyn_radiation` package, rather than changing the existing `radiation` module, so that the legacy stationary path remains available and bitwise comparisons remain possible. The current ADM branch implements spatial transport, angular geodesic advection, radiation-matter coupling, and the radiation energy source in a Valencia-compatible form. The

angular advection coefficients are rebuilt from cached ADM geometry each stage, so ADM mode now supports `angular_fluxes=true`. Radiation is deliberately one-way coupled to the metric in this pass: the metric can affect radiation through the ADM fields, but radiation does not source the Z4c evolution. The implementation remains separate from the legacy solver so stationary CKS behavior can still be tested directly.

The paper is organized as follows. Section 2 fixes notation and derives the radiation variables from the invariant distribution function. Section 3 summarizes the stationary-metric solver. Section 4 derives the ADM variables, geometric source terms, and angular geodesic operator used by the new solver. Section 5 describes the numerical algorithm and implementation. Section 6 documents the null-particle beam-edge diagnostic. Section 7 presents the regression and smoke tests. Section 9 summarizes direct consistency checks. Section 10 summarizes limitations and next implementation steps.

2 Conventions and Radiation Variables

We use units with $c = G = 1$ and metric signature $(-+++)$. Greek indices $\mu, \nu, \dots$ run over spacetime components 0, 1, 2, 3, Latin indices $i, j, \dots$ run over spatial components 1, 2, 3, and hatted Latin indices $\hat{a}, \hat{b}, \dots$ denote an orthonormal spatial tetrad. Repeated indices are summed. The radiation module is grey: it evolves intensity integrated over photon frequency, but still resolves propagation direction on a discrete angular grid.

The invariant starting point is the photon distribution function $f(x^\mu, p_i)$ on the null mass shell. In vacuum, Liouville's theorem says that f is constant along null geodesics:

$$\frac{df}{d\lambda} = p^\mu \frac{\partial f}{\partial x^\mu} + \frac{dp_i}{d\lambda} \frac{\partial f}{\partial p_i} = 0, \quad (1)$$

where λ is an affine parameter and p_μ is the photon covector. Matter interaction adds an invariant collision term to the right-hand side. For a local orthonormal observer measuring photon frequency ν , the frequency-dependent specific intensity satisfies

$$\frac{I_\nu}{\nu^3} \propto f. \quad (2)$$

After integrating over frequency, the bolometric grey intensity transforms as

$$I \propto \nu^4 f. \quad (3)$$

This is why the matter-coupling kernel converts Eulerian-frame intensity to fluid-comoving intensity with a fourth power of the frequency ratio.

The angular grid stores a finite set of unit directions $\ell_n^{(\hat{a})}$ and quadrature weights w_n , with

$$\ell_n^{(\hat{a})} \ell_{n(\hat{a})} = 1, \quad \sum_n w_n \simeq 4\pi. \quad (4)$$

The continuum angular integrals are approximated by these quadrature sums. The code variable U_n is not the invariant distribution function itself; it is a finite-volume conserved density whose precise normalization depends on the metric formulation. The CKS compatibility path keeps the historical Killing-energy normalization. The ADM path uses the Eulerian volume density $U_n = \sqrt{\gamma} I_n$, derived below.

3 Stationary-Metric Radiation Solver

The existing AthenaK radiation package evolves grey specific intensity on a set of angular bins. For each angular bin n , a local orthonormal tetrad direction is

$$q_n^{(\hat{\alpha})} = (1, \ell_n, m_n, p_n), \quad \ell_n^2 + m_n^2 + p_n^2 = 1. \quad (5)$$

The stationary Cartesian Kerr-Schild (CKS) tetrad maps this direction to a coordinate null vector,

$$k_n^\mu = e_{(\hat{\alpha})}^\mu q_n^{(\hat{\alpha})}, \quad k_{\mu,n} = g_{\mu\nu} k_n^\nu. \quad (6)$$

For a stationary metric, k_0 is the photon Killing energy. The legacy finite-volume variable is therefore

$$U_n = k_n^0 k_{0,n} I_n, \quad (7)$$

and the reconstructed primitive entering the spatial flux is

$$k_{0,n} I_n = \frac{U_n}{k_n^0}. \quad (8)$$

The corresponding spatial flux in direction i is

$$F_n^i = k_n^i k_{0,n} I_n. \quad (9)$$

Ignoring matter coupling for the moment, the schematic update is

$$\partial_t U_n + \partial_i F_n^i + \nabla_\Omega \cdot A_n = S_n, \quad (10)$$

where A_n is the angular-space flux induced by geodesic bending. In the stationary CKS module the tetrad, angular face maps, and angular drift coefficients are precomputed. This is efficient and accurate on a fixed black hole background, but it encodes the assumptions that the metric is known analytically, stationary, and tied to the CKS tetrad construction.

4 ADM Formulation for Dynamical Backgrounds

4.1 ADM split and Eulerian observers

The dynamical branch uses the standard ADM line element

$$ds^2 = -\alpha^2 dt^2 + \gamma_{ij} (dx^i + \beta^i dt)(dx^j + \beta^j dt), \quad (11)$$

where α is the lapse, β^i is the shift, and γ_{ij} is the spatial metric. The Eulerian normal observer has

$$n^\mu = \left(\frac{1}{\alpha}, -\frac{\beta^i}{\alpha} \right), \quad n_\mu = (-\alpha, 0, 0, 0). \quad (12)$$

The determinant of the spatial metric is denoted γ , so the proper spatial volume element is $\sqrt{\gamma} d^3x$.

The radiation tetrad timelike leg is chosen to be the Eulerian observer,

$$e_{(\hat{0})}^\mu = n^\mu, \quad (13)$$

and define spatial tetrad columns $E_{(\hat{a})}^i$ satisfying

$$e_{(\hat{a})}^\mu = (0, E_{(\hat{a})}^i), \quad \gamma_{ij} E_{(\hat{a})}^i E_{(\hat{b})}^j = \delta_{\hat{a}\hat{b}}. \quad (14)$$

The dual spatial co-triad is $e_i^{(\hat{a})}$, with

$$e_i^{(\hat{a})} E_{(\hat{b})}^i = \delta_{\hat{b}}^{\hat{a}}, \quad \gamma_{ij} = e_i^{(\hat{a})} e_{(\hat{a})j}. \quad (15)$$

For angular-bin direction $\ell_n^{(\hat{a})}$, the Eulerian spatial unit vector in coordinates is

$$s_n^i = E_{(\hat{a})}^i \ell_n^{(\hat{a})}, \quad \gamma_{ij} s_n^i s_n^j = 1. \quad (16)$$

4.2 Densitized intensity

For a photon with Eulerian frequency

$$\nu_E = -n_\mu p^\mu = \alpha p^0, \quad (17)$$

write the photon four-momentum as

$$p^\mu = \nu_E (n^\mu + s^\mu), \quad s^\mu = (0, s^i), \quad (18)$$

where $s^\mu n_\mu = 0$ and $\gamma_{ij} s^i s^j = 1$. Equation (18) immediately gives the coordinate propagation speed

$$v_n^i = \frac{p^i}{p^0} = \alpha s_n^i - \beta^i. \quad (19)$$

The ADM conserved variable is the Eulerian proper-volume density of grey intensity,

$$U_n = \sqrt{\gamma} I_n, \quad (20)$$

with spatial flux

$$F_n^i = \sqrt{\gamma} v_n^i I_n. \quad (21)$$

Equations (19)–(21) reduce to ordinary flat-space transport when $\alpha = 1$, $\beta^i = 0$, and $\gamma_{ij} = \delta_{ij}$. They also make the normalization difference from the legacy solver explicit: ADM mode evolves the volume-weighted intensity seen by Eulerian observers, not the Killing-energy weighted intensity in Equation (7).

4.3 Radiation moments and Valencia equations

The stress-energy tensor of a grey radiation field can be decomposed relative to the Eulerian observer as

$$T_{\text{rad}}^{\mu\nu} = E n^\mu n^\nu + n^\mu F^\nu + F^\mu n^\nu + P^{\mu\nu}, \quad (22)$$

where $F^\mu n_\mu = 0$ and $P^{\mu\nu} n_\nu = 0$. In the tetrad quadrature used by the code,

$$E = \sum_n w_n I_n, \quad (23)$$

$$F^i = \sum_n w_n I_n s_n^i, \quad (24)$$

$$P^{ij} = \sum_n w_n I_n s_n^i s_n^j, \quad (25)$$

and $P_j^i = P^{ik} \gamma_{kj}$. These definitions are useful for verification: summing the discrete per-angle equations must recover the conservative equations for E and $S_i = \gamma_{ij} F^j$.

Starting from stress-energy conservation,

$$\nabla_\mu T_{\text{rad}}^{\mu\nu} = G^\nu, \quad (26)$$

where G^ν is the radiation four-force density, the Eulerian energy equation is obtained by projecting along n_ν . In vacuum ($G^\nu = 0$), or before applying matter coupling, this gives

$$\partial_t(\sqrt{\gamma}E) + \partial_i [\sqrt{\gamma}(\alpha F^i - \beta^i E)] = \sqrt{\gamma} (\alpha P^{ij} K_{ij} - F^i \partial_i \alpha). \quad (27)$$

Projecting Equation (26) into the spatial slice gives the momentum equation

$$\partial_t(\sqrt{\gamma}S_j) + \partial_i [\sqrt{\gamma}(\alpha P_j^i - \beta^i S_j)] = \sqrt{\gamma} \left[\frac{1}{2} \alpha P^{ik} \partial_j \gamma_{ik} + S_i \partial_j \beta^i - E \partial_j \alpha \right]. \quad (28)$$

Equations (27) and (28) are the same geometric source structure used by Valencia GR hydrodynamics. The only time derivative of the spatial metric that enters is through the extrinsic curvature K_{ij} . This is the central reason the ADM radiation path is not tied to a stationary gauge.

For a single angular bin, substitute

$$E \rightarrow I_n, \quad F^i \rightarrow I_n s_n^i, \quad P^{ij} \rightarrow I_n s_n^i s_n^j.$$

The per-angle geometric energy source used in the code is therefore

$$S_n^{\text{geom}} = U_n (\alpha K_{ij} s_n^i s_n^j - s_n^i \partial_i \alpha). \quad (29)$$

Summing Equation (29) over angles reproduces the right-hand side of Equation (27). The dependence on metric time evolution is carried by K_{ij} . Thus the source is agnostic to the gauge condition used to obtain the ADM data and does not require assuming $\partial_t \alpha = 0$, $\partial_t \beta^i = 0$, or $\partial_t \gamma_{ij} = 0$.

4.4 ADM angular geodesic advection

The angular advection operator accounts for geodesic bending, i.e. the change of $\ell^{(\hat{a})}$ as a ray moves through a curved or time-dependent tetrad. It is derived from the null Hamiltonian. The inverse ADM metric is

$$g^{00} = -\frac{1}{\alpha^2}, \quad g^{0i} = \frac{\beta^i}{\alpha^2}, \quad g^{ij} = \gamma^{ij} - \frac{\beta^i \beta^j}{\alpha^2}. \quad (30)$$

The null constraint $g^{\mu\nu} p_\mu p_\nu = 0$ gives

$$-p_0 = H(x^i, p_i) = \alpha \Pi - \beta^i p_i, \quad \Pi = \sqrt{\gamma^{ij} p_i p_j}. \quad (31)$$

Using coordinate time t as the evolution parameter, Hamilton's equations are

$$\dot{x}^i = \frac{\partial H}{\partial p_i} = \alpha \frac{\gamma^{ij} p_j}{\Pi} - \beta^i, \quad (32)$$

$$\dot{p}_i = -\frac{\partial H}{\partial x^i} = -(\partial_i \alpha) \Pi + p_j \partial_i \beta^j - \frac{\alpha}{2\Pi} p_j p_k \partial_i \gamma^{jk}. \quad (33)$$

The code normalizes the momentum by the Eulerian photon energy, so $\Pi = 1$,

$$p_i = e_i^{(\hat{a})} \ell_{(\hat{a})}, \quad s^i = \gamma^{ij} p_j = E^i_{(\hat{a})} \ell^{(\hat{a})}. \quad (34)$$

With this normalization, Equations (31) and (33) reduce to the implementation formulas

$$H(x^i, p_i) = \alpha \sqrt{\gamma^{ij} p_i p_j} - \beta^i p_i. \quad (35)$$

For the Eulerian tetrad normalization used here, $\sqrt{\gamma^{ij} p_i p_j} = 1$, $p_i = e^{(\hat{a})}_i \ell_{(\hat{a})}$, and

$$\dot{x}^i = \alpha s^i - \beta^i, \quad (36)$$

$$\dot{p}_i = -\partial_i \alpha + p_j \partial_i \beta^j - \frac{\alpha}{2} p_j p_k \partial_i \gamma^{jk}. \quad (37)$$

The angular direction is obtained from the co-momentum through

$$\ell^{(\hat{a})} = E^i_{(\hat{a})} p_i. \quad (38)$$

Differentiating in the same triad gauge gives

$$\dot{\ell}^{(\hat{a})} = E^i_{(\hat{a})} \left[\dot{p}_i - \left(\partial_t e^{(\hat{b})}_i + \dot{x}^j \partial_j e^{(\hat{b})}_i \right) \ell_{(\hat{b})} \right]. \quad (39)$$

Any component of $\dot{\ell}^{(\hat{a})}$ parallel to $\ell^{(\hat{a})}$ corresponds to a roundoff-level change in normalization, not a physical angular drift on the unit sphere, and is projected away:

$$\dot{\ell}^{(\hat{a})} \leftarrow \dot{\ell}^{(\hat{a})} - (\ell_{\hat{b}} \dot{\ell}^{(\hat{b})}) \ell^{(\hat{a})}. \quad (40)$$

The time derivative of the co-triad is obtained by differentiating the same Cholesky factorization used for the spatial triad, with

$$\partial_t \gamma_{ij} = -2\alpha K_{ij} + D_i \beta_j + D_j \beta_i. \quad (41)$$

This relation is an identity in the ADM split and is independent of the particular gauge condition used to evolve α and β^i .

4.5 Cholesky tetrad gauge and derivatives

The implementation uses a Cholesky gauge for the spatial tetrad. Let $\gamma_{ij} = L_{i\hat{a}} L_{j\hat{a}}$, with L lower triangular:

$$L_{1\hat{1}} = \sqrt{\gamma_{11}}, \quad L_{2\hat{1}} = \frac{\gamma_{12}}{L_{1\hat{1}}}, \quad L_{3\hat{1}} = \frac{\gamma_{13}}{L_{1\hat{1}}}, \quad (42)$$

$$L_{2\hat{2}} = \sqrt{\gamma_{22} - L_{2\hat{1}}^2}, \quad L_{3\hat{2}} = \frac{\gamma_{23} - L_{3\hat{1}} L_{2\hat{1}}}{L_{2\hat{2}}}, \quad L_{3\hat{3}} = \sqrt{\gamma_{33} - L_{3\hat{1}}^2 - L_{3\hat{2}}^2}. \quad (43)$$

The co-triad is $e_i^{(\hat{a})} = L_{i\hat{a}}$, and the triad columns are $E^i_{(\hat{a})} = (L^{-T})^i_{\hat{a}}$. For any derivative operator ∂_q , where q may be a spatial coordinate or t , the code uses the differentiated Cholesky relations

$$\partial_q L_{1\hat{1}} = \frac{\partial_q \gamma_{11}}{2L_{1\hat{1}}}, \quad (44)$$

$$\partial_q L_{2\hat{1}} = \frac{\partial_q \gamma_{12} - L_{2\hat{1}} \partial_q L_{1\hat{1}}}{L_{1\hat{1}}}, \quad (45)$$

$$\partial_q L_{3\hat{1}} = \frac{\partial_q \gamma_{13} - L_{3\hat{1}} \partial_q L_{1\hat{1}}}{L_{1\hat{1}}}, \quad (46)$$

$$\partial_q L_{2\hat{2}} = \frac{\partial_q \gamma_{22} - 2L_{2\hat{1}} \partial_q L_{2\hat{1}}}{2L_{2\hat{2}}}, \quad (47)$$

$$\partial_q L_{3\hat{2}} = \frac{\partial_q \gamma_{23} - (\partial_q L_{3\hat{1}}) L_{2\hat{1}} - L_{3\hat{1}} (\partial_q L_{2\hat{1}}) - L_{3\hat{2}} (\partial_q L_{2\hat{2}})}{L_{2\hat{2}}}, \quad (48)$$

$$\partial_q L_{3\hat{3}} = \frac{\partial_q \gamma_{33} - 2L_{3\hat{1}} \partial_q L_{3\hat{1}} - 2L_{3\hat{2}} \partial_q L_{3\hat{2}}}{2L_{3\hat{3}}}. \quad (49)$$

These equations are important for reproducibility: the angular drift uses the same tetrad gauge as the spatial transport and matter-coupling transforms.

4.6 Projection to the angular finite-volume grid

The geodesic angular mesh is a finite-volume mesh on the unit sphere. Writing

$$\ell = (\sin \theta \cos \psi, \sin \theta \sin \psi, \cos \theta),$$

the tangent angular velocities are

$$\dot{\theta} = -\frac{\dot{\ell}_z}{\sin \theta}, \quad (50)$$

$$\sin^2 \theta \dot{\psi} = \ell_x \dot{\ell}_y - \ell_y \dot{\ell}_x. \quad (51)$$

The code projects these two components onto the precomputed angular face normal stored by the geodesic grid. The resulting scalar face speed is the array `na`. In ADM mode `na` is already a coordinate-time angular velocity, so the angular finite-volume flux upwinds U_n directly. In the CKS compatibility branch the historical angular primitive U_n/k_n^0 is still used, preserving the stationary solver behavior.

5 Numerical Algorithm and Implementation

5.1 Standalone solver layout

The new module is intentionally independent of `src/radiation`. It lives under `src/dyn_radiation` and supports two metric modes:

CKS compatibility mode uses the same normalization and transport variables as the legacy solver, enabling direct regression comparisons.

ADM mode uses Equations (20)–(29) and obtains α , β^i , γ_{ij} , and K_{ij} from the active relativistic mesh data.

The solver is constructed as a relativistic component when either the CKS path or the dynamical ADM path is active. This is required so that the radiation state has access to the metric-dependent arrays used by the flux and source kernels.

For Z4c evolutions the present solver is a test-field radiation system. The NR task graph no longer queues a radiation stress-energy task before `Z4c.CalcRHS`, and `Tmunu` storage is not allocated solely because `dyn_radiation` is enabled. Matter stress-energy from the MHD/dynGRMHD path is unchanged. A run that combines `dyn_radiation` and `z4c` must use `geometry=adm`; the CKS mode is retained only for fixed background compatibility and regression tests.

5.2 Finite-volume equation actually advanced

For each angular cell n , ADM mode advances

$$\partial_t U_n + \partial_i (\sqrt{\gamma} v_n^i I_n) + \mathcal{D}_\Omega (U_n \ell_n) = S_n^{\text{geom}} + C_n, \quad (52)$$

where $U_n = \sqrt{\gamma} I_n$, $v_n^i = \alpha s_n^i - \beta^i$, $\mathcal{D}_\Omega$ is the finite-volume angular divergence on the geodesic grid, S_n^{geom} is Equation (29), and C_n contains emission, absorption, scattering, and user source terms. On a Cartesian mesh cell with coordinate widths Δx^i , the explicit transport part of one Runge–Kutta substep is

$$U_n^* = U_n^m - \Delta t \sum_i \frac{F_{n,+}^i - F_{n,-}^i}{\Delta x^i} - \Delta t \frac{1}{\Delta \Omega_n} \sum_{f \in \partial \Omega_n} \Delta l_f A_{n,f}, \quad (53)$$

where $F_{n,\pm}^i$ are reconstructed spatial face fluxes, $\Delta \Omega_n = w_n$, Δl_f is the angular edge length, and

$$A_{n,f} = n_{n,f}^a \begin{cases} U_{\text{upwind}}, & \text{ADM mode,} \\ (U/k^0)_{\text{upwind}}, & \text{CKS compatibility mode.} \end{cases} \quad (54)$$

Here $n_{n,f}^a$ is the scalar angular face speed stored in `na`. The distinction in Equation (54) is the main normalization difference between the two metric modes.

5.3 Stage update

At each explicit stage, ADM mode performs the following operations:

1. Refresh ADM variables on cell centers and faces from the stage-start geometry.
2. Construct Eulerian spatial tetrads by factorizing γ_{ij} .
3. Fill cached $\sqrt{\gamma}$, α , β^i , γ_{ij} , γ^{ij} , K_{ij} , tetrad columns, co-triads, spatial metric gradients, and co-triad derivatives.
4. Reconstruct $I_n = U_n / \sqrt{\gamma}$ to faces.
5. Compute face speeds $v_n^i = \alpha s_n^i - \beta^i$ and fluxes from Equation (21).
6. Compute angular drift from Equations (37) and (39); form angular fluxes using U_n as the upwinded quantity.
7. Apply flux divergence to U_n .
8. Apply the per-angle geometric source in Equation (29).

9. Apply matter/source terms, boundary exchange, and AMR operations.

In the NR task graph this refresh is a dedicated `Rad_PrepGeom` task. `Rad_Flux` and the radiation coupling path depend on this cache refresh and therefore use the same stage-start ADM state. `Rad_Flux` does not wait for the post-update `Z4c_Z4c2ADM` conversion. The radiation timestep task may wait for `Z4c_Z4c2ADM`, so the next timestep is limited by the post-stage ADM geometry. This ordering keeps radiation transport consistent with the RK stage while still measuring the next CFL limit on the latest metric state.

The cache is deliberately cell-local and device-resident. The primary state arrays store the ADM fields and co-triad:

```
adm_alpha_c,  adm_beta_u_c,  adm_g_dd_c,
adm_g_uu_c,  adm_K_dd_c,  adm_cotriad_c.
```

The derivative arrays

```
adm_grad_alpha_c,  adm_grad_beta_u_c,
adm_grad_g_dd_c,  adm_grad_g_uu_c,  adm_grad_cotriad_c
```

store centered spatial derivatives, using one-sided differences only at the edge of the allocated block. The array `adm_dt_cotriad_c` stores $\partial_t e_i^{(\hat{a})}$ computed from Equations (41) and (49). These arrays are Kokkos device views; no host containers or dynamic allocation enter the transport kernels.

The ADM CFL limiter has three parts: coordinate spatial speeds, angular-geodesic speeds, and the explicit geometric source estimate. In runs with radiation excision, all three reductions skip cells marked by `rad_mask`; otherwise masked cells with singular or intentionally unphysical metric data can set a meaningless global timestep.

5.4 Beam source and output normalization

The beam source now uses the active radiation tetrad, rather than assuming the legacy CKS metric, to construct the beam angle and source normalization. The increment to the conserved radiation variable is

$$\Delta U_n = \begin{cases} k_n^0 k_{0,n} \dot{I} \Delta t, & \text{legacy or CKS-compatible normalization,} \\ \sqrt{\gamma} \dot{I} \Delta t, & \text{ADM normalization.} \end{cases} \quad (55)$$

Moment output applies the inverse normalization,

$$I_n = \begin{cases} U_n / (k_n^0 k_{0,n}), & \text{legacy or CKS-compatible normalization,} \\ U_n / \sqrt{\gamma}, & \text{ADM normalization.} \end{cases} \quad (56)$$

This separation is what allows the flat ADM beam test to reproduce the legacy flat CKS beam component by component, even though the conserved variables differ internally.

5.5 Radiation-matter coupling in ADM variables

The matter-coupling routine is local to a cell and follows the legacy grey implicit update, but with ADM frame transformations and moment feedback. The fluid primitive velocity variables are the Valencia spatial four-velocity components $u_{(\text{spatial})}^i$. The Lorentz factor measured by the Eulerian observer is

$$W = \sqrt{1 + \gamma_{ij} u_{(\text{spatial})}^i u_{(\text{spatial})}^j}. \quad (57)$$

The ADM normal-to-tetrad transform maps the fluid four-velocity into the Eulerian orthonormal frame:

$$u^{(\hat{0})} = W, \quad u^{(\hat{a})} = e^{(\hat{a})}_i u^i_{(\text{spatial})}. \quad (58)$$

For a photon direction $(1, \ell^{(\hat{a})})$ in the Eulerian tetrad, the frequency ratio between the fluid-comoving frame and the Eulerian frame is

$$D \equiv \frac{\nu_{\text{cm}}}{\nu_{\text{E}}} = u^{(\hat{0})} - u^{(\hat{a})} \ell_{(\hat{a})}. \quad (59)$$

This is the special-relativistic Doppler factor written in the local Eulerian tetrad. The grey intensity and solid angle used by the coupling update are

$$I_{\text{cm}} = I_{\text{E}} D^4, \quad d\Omega_{\text{cm}} = D^{-2} d\Omega_{\text{E}}, \quad (60)$$

which follows from I_ν/ν^3 invariance and aberration. In ADM mode $I_{\text{E}} = U/\sqrt{\gamma}$. In CKS compatibility mode the legacy normalization $I_{\text{E}} = U/(k^0 k_0)$ is retained.

The grey absorption/scattering update can be written schematically as

$$\frac{\Delta I_{\text{cm}}}{\Delta \tau} = \sigma_a (B - I_{\text{cm}}) + \sigma_s (J_{\text{cm}} - I_{\text{cm}}) + \sigma_p (B - J_{\text{cm}}), \quad (61)$$

where $B = a_{\text{rad}} T_{\text{gas}}^4 / (4\pi)$, J_{cm} is the comoving-frame angular mean intensity, and $\sigma_a, \sigma_s, \sigma_p$ are the absorption, scattering, and Planck-correction coefficients used by the existing opacity helper. The gas internal energy is linear in T_{gas} for the ideal-gas EOS, while $B \propto T_{\text{gas}}^4$. The local implicit solve therefore reduces to the same quartic temperature equation used by the legacy solver. The new ADM-specific parts are the frame normalization in Equations (59)–(60) and the feedback moments below.

Before and after the implicit radiation update, ADM mode computes the Valencia radiation conserved moments

$$\sqrt{\gamma} E = \sum_n w_n U_n, \quad (62)$$

$$\sqrt{\gamma} S_i = \sum_n w_n U_n \gamma_{ij} s_n^j. \quad (63)$$

The gas conserved variables receive the opposite change,

$$\Delta(\sqrt{\gamma} \tau_{\text{gas}}) = -\Delta(\sqrt{\gamma} E_{\text{rad}}), \quad \Delta(\sqrt{\gamma} S_i^{\text{gas}}) = -\Delta(\sqrt{\gamma} S_i^{\text{rad}}), \quad (64)$$

up to the existing choice of whether `affect_fluid` is enabled. This is the practical consistency condition for local matter coupling: the same Eulerian radiation variables used in the transport equation are used to feed energy and momentum back to the fluid.

5.6 Cost of tetrad construction

In the existing stationary solver, tetrads and angular maps are generated once. For a dynamical spacetime, reconstructing a full tetrad for every angle would be too expensive. The present ADM implementation avoids that cost by constructing only the three spatial tetrad columns per cell or face. Every angular bin then forms

$$s_n^i = E_{(\hat{a})}^i \ell_n^{(\hat{a})} \quad (65)$$

with a small dense matrix-vector product. The natural next optimization is to cache all metric quantities needed by Equations (37) and (39) once per stage, then reuse them across all angular transport and source kernels. That cache now contains $\sqrt{\gamma}$, α , β^i , γ_{ij} , γ^{ij} , K_{ij} , the triad and co-triad, spatial gradients, and $\partial_t e_i^{(\hat{a})}$. This keeps the cost scaling as $O(N_{\text{cell}})$ for metric preparation plus $O(N_{\text{cell}}N_{\Omega})$ for angular transport, instead of rebuilding a complete tetrad inside each angle loop.

5.7 Code-to-equation map

Table 1 maps the main ADM formulas to the implementation. The purpose is to make review of signs and normalizations straightforward.

Mathematical object	Code storage or routine	Main use
α	<code>adm.alpha.c</code>	speed, source, Hamiltonian
β^i	<code>adm.beta.u.c</code>	$v^i = \alpha s^i - \beta^i$, Hamiltonian
γ_{ij}, γ^{ij}	<code>adm.g_dd.c</code> , <code>adm.g_uu.c</code>	tetrad, momentum, metric gradients
K_{ij}	<code>adm.K_dd.c</code>	Valencia energy source
$E_i^{(\hat{a})}$	<code>tet.c</code> spatial block	coordinate direction s^i
$e_i^{(\hat{a})}$	<code>adm.cotriad.c</code>	momentum p_i , fluid tetrad velocity
$\partial_i \alpha, \partial_i \beta^j, \partial_i \gamma^{jk}$	<code>adm.grad.*</code> arrays	Hamiltonian $\dot{p}_i$, energy source
$\partial_t e_i^{(\hat{a})}$	<code>adm.dt_cotriad.c</code>	tetrad time-dependence in $\ell^{(\hat{a})}$
$\hat{\ell} \cdot \hat{n}_{\Omega}$	<code>na</code>	angular finite-volume flux

Table 1: Code-to-equation map for the ADM radiation implementation.

6 Null Geodesic Beam-Edge Tracers

The particle module now has an opt-in `null_geodesic` pusher for diagnostic beam-edge trajectories on ADM backgrounds. Particle real data are interpreted as

$$(x, y, z, v_x, v_y, v_z) \longrightarrow (x^i, k_i), \quad (66)$$

where k_i is the covariant spatial photon momentum. The coordinate-time Hamiltonian is

$$\mathcal{H}(x^i, k_i) = \alpha \sqrt{\gamma^{ij} k_i k_j} - \beta^i k_i. \quad (67)$$

Let

$$K = \sqrt{\gamma^{ij} k_i k_j}. \quad (68)$$

The pusher advances

$$\frac{dx^i}{dt} = \frac{\partial \mathcal{H}}{\partial k_i} = \alpha \frac{\gamma^{ij} k_j}{K} - \beta^i, \quad (69)$$

$$\frac{dk_i}{dt} = -\frac{\partial \mathcal{H}}{\partial x^i} = -(\partial_i \alpha) K - \frac{\alpha}{2K} (\partial_i \gamma^{jk}) k_j k_k + (\partial_i \beta^j) k_j. \quad (70)$$

The current implementation uses the nearest cell's ADM fields and centered finite differences for the metric derivatives. That accuracy is sufficient for beam-edge diagnostics and visualization. Higher-order interpolation can be added later without changing the particle data layout.

7 Verification and Regression Tests

7.1 Stationary and flat-space beam tests

The first regression compares three beam calculations:

1. the legacy `radiation` solver on a flat CKS background,
2. `dyn_radiation` in CKS compatibility mode, and
3. `dyn_radiation` in ADM flat-space mode.

The reproducibility inputs are stored in

```
dyngr_radiation_paper/inputs/beam_radiation_cks.athinput
dyngr_radiation_paper/inputs/beam_dynrad_cks.athinput
dyngr_radiation_paper/inputs/beam_dynrad_adm_flat.athinput
```

The maximum absolute difference in R^{00} between the legacy CKS beam and the standalone CKS beam is zero in the current regression. The maximum absolute difference between the legacy CKS beam and the ADM flat-space beam is also zero. In addition to the one-dimensional slice in Figure 1, the same runs now write a two-dimensional x - y binary slice at $z = 0$. The two-dimensional fields are shown in Figure 2, and residual maps against the legacy solver are shown in Figure 3. The maximum absolute residuals in the 2D maps are 0 for both the CKS-compatible and ADM-flat branches. The same ADM flat beam input also completed under `mpirun -n 2`, testing MPI decomposition for this path.

A separate stationary Kerr-Schild ADM smoke test is included at `inputs/tests/dynrad_bh_beam_adm.athinput`. That input keeps the metric time-independent but non-flat, and runs `geometry=adm` with `angular_fluxes=true`. It therefore exercises the Hamiltonian angular drift, metric-gradient cache, and ADM angular timestep limiter without relying on the BBH problem generator. The short regression completed to $t = 0.2$ with finite radiation output. The excision-aware ADM CFL limiter kept the minimum reported timestep at 4.374670×10^{-2} , rather than letting masked black-hole cells dominate the reduction.

The dynamic-metric smoke input

```
inputs/tests/dynrad_z4c_wave_adm.athinput
```

adds `dyn_radiation` to the Z4c linear-wave background. This checks that ADM metric fields supplied by the Z4c-to-ADM conversion are accepted by the standalone radiation module and that the per-stage geometry cache can be refreshed on a time-dependent metric with `angular_fluxes=true`. The compact regression completed without NaNs or infinities.

The tracked stress harness `dyngr_radiation_paper/scripts/run_stress_tests.py` runs the compact serial, restart, MPI, and dynbbh cases with outputs under `/tmp`. In the current run it passed all 20 checks: eight serial solver inputs, five two-rank MPI inputs, the write/resume restart pair, the expected fatal rejection of `dyn_radiation geometry=cks` with `z4c`, the expected dynbbh rejection of legacy `radiation`, two serial dynbbh inputs, and one two-rank dynbbh smoke test. The harness scans text outputs and logs for non-finite values.

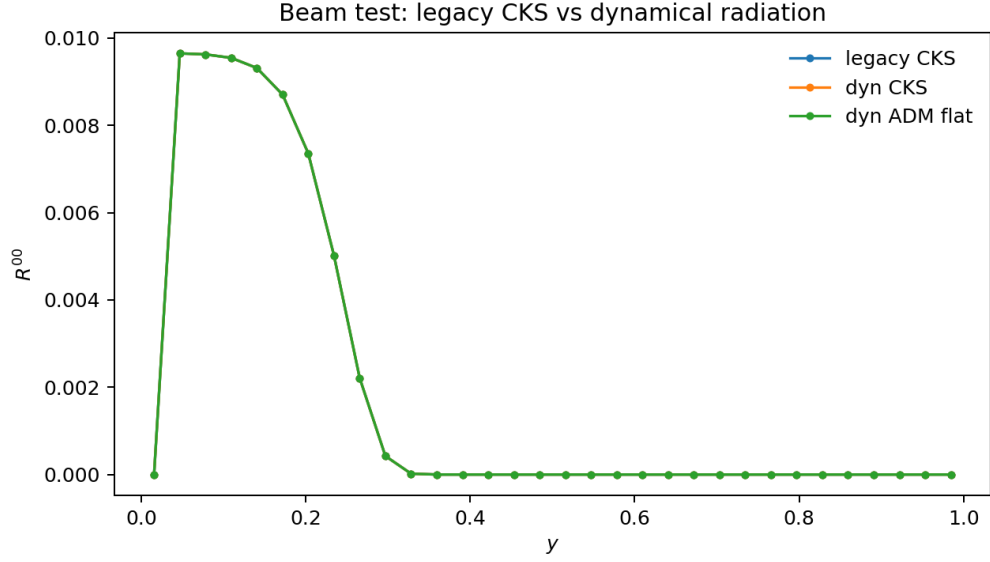


Figure 1: Radiation moment R^{00} along the output slice for the legacy CKS solver, the standalone CKS-compatible branch, and the ADM flat-space branch. The plotted profiles overlap in the current regression.

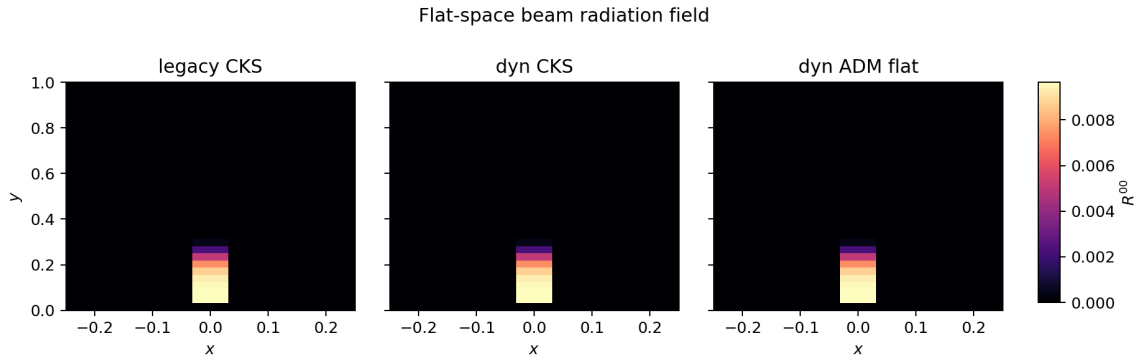


Figure 2: Two-dimensional R^{00} fields for the flat-space beam test. The legacy CKS solver, the standalone CKS-compatible branch, and the ADM flat-space branch use the same color scale.

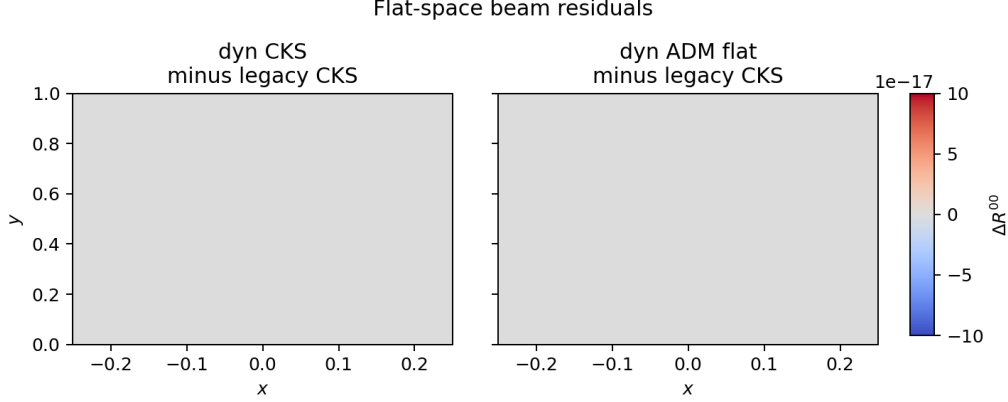


Figure 3: Two-dimensional residuals in R^{00} for the flat-space beam test, computed against the legacy CKS solution. The residuals are zero to the reported precision in the current regression.

7.2 Radiation-hydrodynamic linear wave

The stationary CKS compatibility branch was also checked against the legacy radiation solver on a radiation-hydrodynamic linear wave. The final error rows match:

$$\begin{aligned}
 \text{RMS-}L_1 &= 2.400907 \times 10^{-6}, \\
 d_{L_1} &= 1.685676 \times 10^{-6}, \\
 M_{L_1}^1 &= 1.380892 \times 10^{-6}, \\
 E_{L_1} &= 1.007962 \times 10^{-6}.
 \end{aligned} \tag{71}$$

The corresponding inputs are `dyngr_radiation_paper/inputs/lwave_radiation_cks.athinput` and `dyngr_radiation_paper/inputs/lwave_dynrad_cks.athinput`.

7.3 Binary-background beam and particle smoke test

The binary-background input is

`dyngr_radiation_paper/inputs/dynbbh_beam_particles.athinput`.

It exercises the ADM beam source on the superposed boosted Kerr-Schild binary background in `src/pgen/dynbbh.cpp`. It runs with `geometry=adm` and `angular_fluxes=true`. The `dynbbh` initializer now uses only `pdynrad`; the legacy `prad` path is rejected because its stationary CKS normalization is inconsistent with the ADM background. It also initializes null particles on the beam edge and advances them using Equations (69) and (70). The serial smoke test produced a finite beam slice with $\max R^{00} = 0.768586$, wrote a normal-size particle track file, and generated the diagnostic figure shown in Figure 5. A second serial run enabled ADM radiation-matter coupling with absorption and scattering opacities and completed without NaNs. The two-rank `dynbbh` smoke test also advanced past cycle zero and completed to its cycle limit. These are compact runtime tests; long production `dynbbh`/MHD radiation evolutions remain a separate validation target.

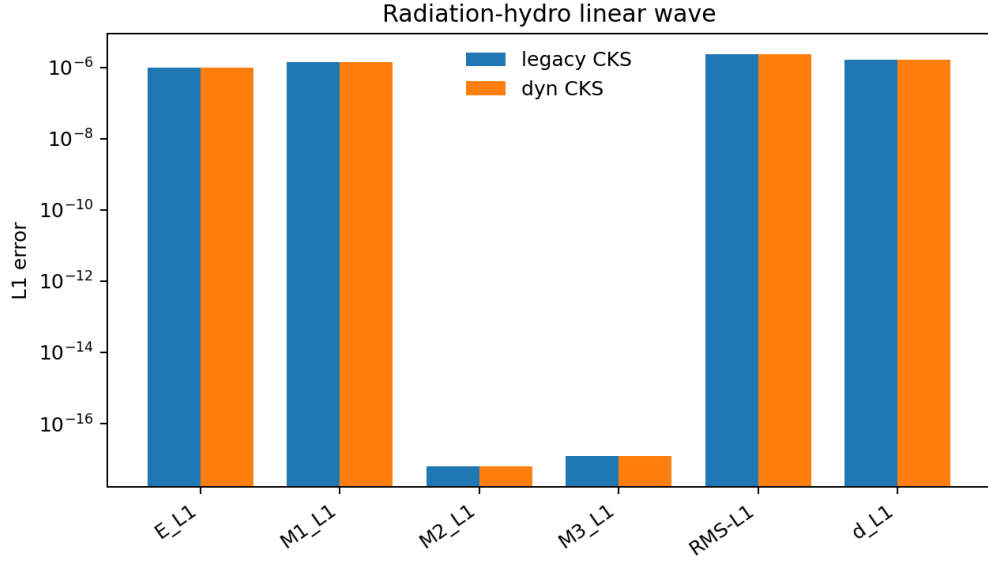


Figure 4: Linear-wave error metrics for the legacy solver and `dyn_radiation` in stationary CKS compatibility mode.

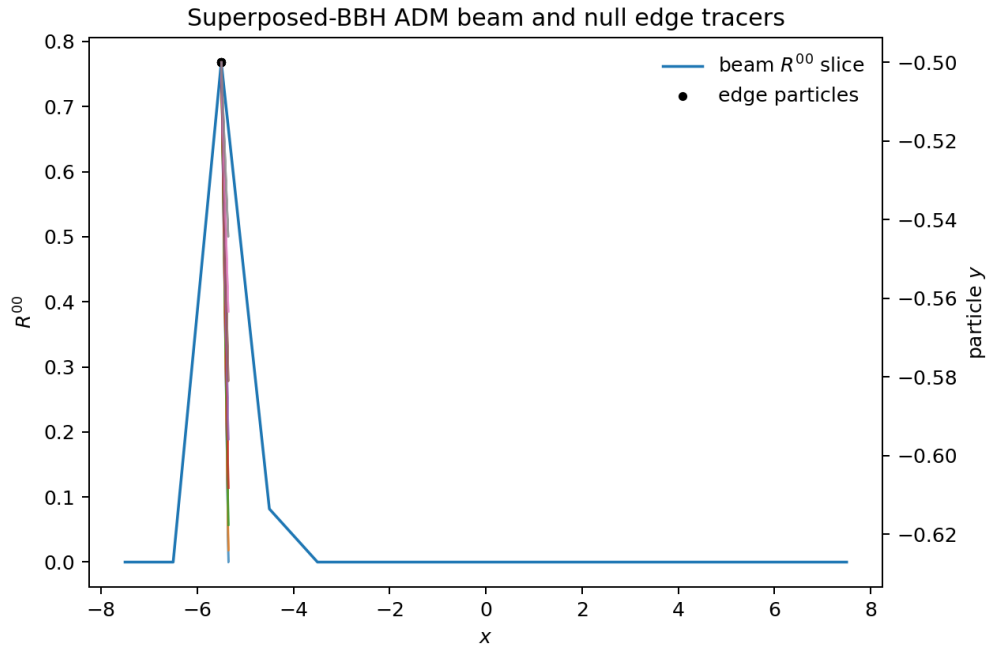


Figure 5: ADM beam slice on the superposed binary black-hole background with null particle trajectories initialized on the edge of the beam cone.

Test	Quantity checked	Result
Flat beam, CKS compatibility	$\max R_{\text{dynrad}}^{00} - R_{\text{legacy}}^{00} $	0
Flat beam, ADM mode	$\max R_{\text{ADM}}^{00} - R_{\text{legacy}}^{00} $	0
Flat ADM beam, MPI	two-rank decomposition	passed
Stationary KS ADM beam	nonzero ADM angular drift	passed
Dynamic Z4c ADM smoke	time-dependent ADM metric cache	passed
CKS linear wave	final error row	identical
Restart	write and resume <code>dynrad_beam_cks</code>	passed
Z4c CKS rejection	unsupported metric mode guard	passed
BBH legacy radiation rejection	<code>prad</code> guard	passed
BBH ADM beam	finite beam and particle tracks	passed in serial
BBH ADM coupling	absorption/scattering with MHD primitives	passed in serial
BBH coupled MPI smoke	two-rank dynbbh/MHD run	passed

Table 2: Current regression status for the standalone dynamical radiation implementation.

Feature	<code>radiation</code>	<code>dyn_radiation</code>
Metric source	Analytic stationary CKS metric	CKS compatibility mode or ADM mesh fields
Conserved intensity	$k^0 k_0 I$	CKS: $k^0 k_0 I$; ADM: $\sqrt{\gamma} I$
Spatial speed	k^i from stationary tetrad	CKS: k^i ; ADM: $\alpha s^i - \beta^i$
Metric source terms	Stationary tetrad and angular drift coefficients	ADM Valencia energy source using K_{ij} and $\partial_i \alpha$
Angular bending	Precomputed CKS coefficients	Cached ADM Hamiltonian angular drift
Matter coupling	Existing local implicit path	CKS path retained; ADM uses $I = U/\sqrt{\gamma}$, Eulerian tetrad frequency, and Valencia moment feedback
Metric backreaction	Not applicable to fixed metric	Not included; radiation is metric-passive
Binary backgrounds	Not supported directly	Supported through ADM fields

Table 3: Functional comparison between the existing stationary radiation solver and the new standalone dynamical-background solver.

8 Comparison With the Existing Solver

The key mathematical difference is the removal of the stationary Killing-energy normalization in ADM mode. The key implementation difference is that metric data are no longer assumed to be analytic or time independent. The new solver therefore fits the same design pattern as the Valencia GRMHD path: fluxes and sources are built from ADM variables supplied by the active spacetime module. At present this analogy is limited to transport and matter coupling. The radiation stress-energy tensor is not passed to the metric solver, so Z4c evolutions see MHD/dynGRMHD matter sources only.

9 How to Check the Formalism

The derivation above gives several concrete checks that can be applied to the implementation.

Flat-space limit. Set $\alpha = 1$, $\beta^i = 0$, $\gamma_{ij} = \delta_{ij}$, and $K_{ij} = 0$. Then $E^i_{(\hat{a})} = \delta^i_{\hat{a}}$, $U_n = I_n$, $v^i = s^i$, $S_n^{\text{geom}} = 0$, and all angular drift coefficients vanish. The ADM solver must reduce to ordinary Cartesian free-streaming. This is the reason the flat ADM beam can match the legacy flat CKS beam exactly.

Moment-source check. For any angular distribution, compute the discrete moments in Equation (25). The angular sum of the per-angle energy source in Equation (29) must equal the right-hand side of Equation (27). A stronger check, still to be automated, is that the angular geodesic operator produces the momentum source in Equation (28) when summed over angles.

Tetrad consistency check. At every cached cell,

$$\gamma_{ij} E^i_{(\hat{a})} E^j_{(\hat{b})} = \delta_{\hat{a}\hat{b}}, \quad e^{(\hat{a})}_i E^i_{(\hat{b})} = \delta^{\hat{a}}_{\hat{b}}. \quad (72)$$

The same Cholesky gauge must be used for spatial transport, angular drift, and matter coupling; otherwise spurious angular drift appears even in locally flat coordinates.

Frequency-transform check. For a fluid comoving with the Eulerian observer, $u^{(\hat{0})} = 1$ and $u^{(\hat{a})} = 0$, so $D = \nu_{\text{cm}}/\nu_{\text{E}} = 1$. The ADM coupling must then use $I_{\text{cm}} = U/\sqrt{\gamma}$. For a moving fluid, the fourth-power intensity transform and the inverse-square solid-angle transform in Equation (60) must be applied together; using one without the other gives the wrong comoving mean intensity.

Local conservation check. When `affect_fluid=true`, the matter-coupling kernel should change gas energy and momentum by the negative of the change in the radiation moments in Equation (63). This is independent of how the implicit temperature solve is performed.

10 Discussion and Limitations

The present implementation is a controlled step toward dynamical radiation transport. It moves radiation energy on ADM backgrounds, bends the angular directions with an ADM Hamiltonian operator, reproduces stationary beam and linear-wave regressions, and diagnoses beam propagation on the superposed binary metric. It is still not a final production radiation module. The current coupling to dynamical spacetime is one-way: ADM/Z4c geometry drives radiation, but radiation

does not feed $T_{\mu\nu}$ back into the Z4c right-hand side. The angular operator uses second-order metric-gradient caches; higher-order gradients and limiters can be added behind the same cache interface. The discrete momentum-source consistency check in Equation (28) is the next validation target.

The ADM matter-coupling path now uses the Eulerian tetrad frequency factor, $I = U/\sqrt{\gamma}$, and gas feedback through the Valencia radiation moments. It has passed flat and BBH smoke tests, but it still needs a broader validation suite with moving-fluid absorption/scattering, optically thick diffusion, and time-dependent metric backgrounds. The explicit ADM update clips negative intensity to zero for positivity; a counter for that floor activation is still needed so production runs can monitor any silent radiation-energy loss.

The changed kernels were reviewed for Kokkos device compatibility: the stage geometry cache, timestep reductions, fluxes, and coupling kernels capture only device views and POD scalars. The tests reported here were run on the local CPU/MPI builds. No CUDA, HIP, or SYCL runtime validation was available in this pass, so GPU execution should be treated as statically reviewed but not runtime-certified.

Finally, the null-particle pusher is currently a diagnostic tool. It uses nearest-cell ADM fields and centered finite differences. It is useful for beam edge visualization and for catching obvious metric or sign errors, but a scientific geodesic integrator should use higher-order metric interpolation, adaptive timestepping, and a stricter monitor of the null Hamiltonian.

11 Conclusions

We implemented and documented a standalone dynamical-background radiation module for AthenaK. The solver preserves legacy CKS behavior in a compatibility mode and adds an ADM mode based on Eulerian densitized intensity, Valencia coordinate speeds, and a per-angle geometric source whose angular sum recovers the Valencia radiation energy equation. The formulation is suitable for time-dependent ADM backgrounds because metric evolution enters through K_{ij} and the ADM co-triad time derivative, rather than through stationarity assumptions. Regression tests show exact agreement with the legacy flat beam, identical stationary CKS linear-wave error norms, successful two-rank flat ADM beam transport, excision-safe stationary black-hole ADM timesteps, restart coverage, and serial plus two-rank ADM beam/null-particle smoke tests on the superposed binary black-hole background. The remaining work is to add radiation metric backreaction, strengthen the momentum-source and matter-coupling validation suite, monitor positivity clipping, and upgrade the diagnostic particle pusher if high-accuracy geodesics are required.